Gauge Structure, QED, and Gravity from the Spectral Geometry of S^3: A Synthesis of the GeoVac QED/Gauge Arc*

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The Geometric Vacuum (GeoVac) framework discretizes the hydrogen bound-state manifold as the unit three-sphere S^3 via Fock's conformal projection. This synthesis assembles the eight Group 5 papers—gauge structure, graph-native QED, precision bound-state QED, and a spectral-action gravity sketch—into one narrative around a single organizing distinction: *the gauge group structure is largely forced by the spectral geometry of S^3 and its Hopf fibration $S^1 \to S^3 \to S^2$, while the couplings are calibration/observation, not derived.* The graph carries a $U(1)$ Wilson/Hodge lattice gauge structure [6], a non-abelian SU(2) Wilson theory because $S^3 = \mathrm{SU}(2)$ as a manifold [8], and—on the cross-ℓ Dirac graph—a $U(1)$ structurally compatible with three-dimensional compact electrodynamics [14]; the natural gauge reach is $U(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, with SU(3) entering only as a pure-gauge theory whose natural matter coupling is obstructed. The framework's QED is genuinely graph-native in its selection-rule and period content: the eight perturbative selection rules partition 1+6+1 by which graph ingredient recovers them [10], and the S^3 Dirac spectral sums reproduce the flat-space QED transcendentals ($\zeta(3)$, $\zeta(5)$, Catalan G, $\zeta(5,3)$) through a depth-filtered period tower [7]. The fine-structure constant satisfies the cubic $\alpha^3 - K\alpha + 1 = 0$ with $K = \pi(B + F - \Delta)$ assembled from three spectral invariants of the Hopf bundle, each with an independently established spectral home; this is recorded strictly as an **Observation** (agreement 8.8×10^{-8}, twelve single-mechanism derivations eliminated), never as a derivation [2]. Bound-state QED reaches sub-percent one-loop Lamb-shift closure with no GeoVac-specific fit [12], and a spectral-action gravity sketch yields an *exactly* two-term Einstein–Hilbert action together with a set of honest negatives that bound a still-open discrete-substrate program [15]. The discipline throughout matches the rest of the corpus: every load-bearing claim wears its tier, the structural results are foregrounded, and the couplings are positioned as the free, calibration side of the construction.

I. INTRODUCTION: FORCED STRUCTURE, FREE COUPLINGS

The GeoVac program establishes that the discrete graph Laplacian for hydrogen is mathematically equivalent to the Laplace–Beltrami operator on the unit three-sphere S^3, related to the Schrödinger equation by Fock's 1935 conformal projection [3]. The Group 5 arc asks what the *same* spectral geometry says about the rest of electromagnetism, the gauge sector, and—most speculatively—gravity. A single distinction organizes the answer, and it is the through-line of this synthesis.

The gauge group structure is largely forced. The three-sphere carries a canonical principal-bundle structure, the Hopf fibration $S^1 \to S^3 \to S^2$, and S^3 *is* the group manifold SU(2). These two facts are not GeoVac inputs; they are geometry. Reading the GeoVac graph through its incidence matrix turns them into an abelian and then a non-abelian Wilson lattice gauge theory on the same nodes and edges that carry the Coulomb spectrum (§II). The natural gauge reach saturates at $U(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$, and where it stops is itself structural: SU(3) admits only a pure-gauge Wilson theory, its natural matter coupling obstructed at the Clebsch–Gordan intertwiner level (a sharpening of the Coulomb/harmonic-oscillator asymmetry of Paper 24 [5]).

The couplings are not. The fine-structure constant, the QED multi-loop constants, and the cutoff function of the gravity sector are all on the other side of the ledger: calibration or observation, in the two-layer language of Paper 34 [11]. The framework's most precise number—α reproduced to 8.8×10^{-8}—is held explicitly at **Observation** tier, with no claim of derivation (§IV). This is the same discipline that runs through the quantum-chemistry and quantum-computing arcs: lead with the structural result, name the calibration boundary, and let the tier of each claim be visible in the prose.

Following the framework's positioning convention, the discrete graph topology and the continuous Hopf/spectral-action geometry are treated as *dual descriptions* connected by proven maps; no ontological priority is asserted for either, and the synthesis leads with concrete structural content rather than interpretive claims about the nature of electromagnetism or gravity.

The eight papers form a loose dependency order. Papers 25, 30, and 41 build the gauge structure. Papers 33 and 28 develop graph-native QED: its selection rules and its perturbative period content. Paper 2 reports the α observation. Paper 36 turns the Dirac-S^3 machinery onto precision bound-state observables. Paper 51 sketches the spectral-action gravity sector and—characteristically—records what did not work.

* Group 5 synthesis in the Geometric Vacuum series

II. GAUGE STRUCTURE FROM THE HOPF BUNDLE

a. The Hodge/Wilson dictionary on the graph (Paper 25). With B the node–edge incidence matrix of the GeoVac graph, the node Laplacian $L_0 = BB^\top$ is the matter propagator (it converges to $-\Delta_{S^3}$ [3]) and the edge Laplacian $L_1 = B^\top B$ is the discrete Hodge-1 Laplacian whose kernel counts cycles—the photon propagator in temporal gauge. Assigning a $U(1)$ connection $U_e = e^{i\theta_e}$ to each edge and collapsing the magnetic label m_ℓ over its $2\ell+1$ Hopf fiber reproduces the S^2 base graph exactly. None of the underlying mathematics is new (graph Hodge theory, Wilson lattice gauge theory, the Marcolli–van Suijlekom gauge-network spectral triple [16]); what Paper 25 claims is the *synthesis*—one physically motivated discretization on which matter, an abelian connection, and the Hopf base coexist. The paper is explicit about its tier: "an observational synthesis paper. No new experimental or computational prediction is derived," and the edge-Laplacian spectrum of a finite Hopf graph is necessarily algebraic (π-free), so no transcendental can hide in the gauge 1-forms themselves.

b. SU(2) *because* $S^3 = \mathrm{SU}(2)$ *(Paper 30).* The rank-1 non-abelian case is special: the manifold *is* the gauge group. With links $U_e \in \mathrm{SU}(2)$, plaquettes the primitive non-backtracking (Ihara) cycles, and Wilson action $S_W = \beta \sum_P \big(1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,U_P\big)$, the construction is a gauge-invariant, Haar-normalizable finite theory. Two results carry real weight. The maximal-torus (Cartan) reduction reproduces Paper 25's $U(1)$ action to machine precision ($|S_W - S_{U(1)}| < 10^{-12}$)—stated as a **proposition with a proof**—so Paper 25 is exactly the abelian sub-sector of a genuinely non-abelian theory. The second-order weak-coupling expansion produces the co-exact (curl) plaquette form $B_2 B_2^\top$ — *not* a multiple of L_1; it is the missing half that completes Paper 25's $L_1 = B^\top B$ to the full discrete Hodge-1 Laplacian (Paper 30 Prop. 2, corrected 2026-07-03) — with non-abelian couplings first entering at order A^4. The kinetic normalization is universal,

$$\text{coeff} = \frac{1}{4N_c} \implies \tfrac{1}{8}\ (\mathrm{SU}(2)), \quad \tfrac{1}{12}\ (\mathrm{SU}(3)), \tag{1}$$

verified symbolically and numerically. The scope is explicitly bounded: this is a compact-graph construction, *not* a Yang–Mills mass-gap or confinement claim on $\mathbb{R}^4$; the Monte-Carlo plaquette data is leading-order and agrees with the character expansion only to the $\sim 20\%$ level at intermediate β.

c. Where the gauge reach stops. The non-abelian story does not continue to SU(3) as a *natural* matter-coupled theory. On the S^5 Bargmann–Segal lattice (the harmonic-oscillator analog of S^3) a pure-gauge Wilson SU(3) is well defined and machine-precision verified (the same $1/(4N_c) = 1/12$ coefficient, Cartan reduction over 7,765 plaquettes to $< 1.5\times10^{-11}$), but the natural matter sector cannot be coupled in the strict Wilson sense: the Clebsch–Gordan intertwiner obstruction re-appears at the matter level, and a least-squares fit of the m_ℓ-quotient to $\mathbb{CP}^2$ fails uniformly (40.8% least-squares residual, with a convention-robust $\geq 33\%$ floor; Paper 25 §VII). This is the gauge-side face of the Coulomb/HO asymmetry catalogued in Paper 24 [5]: $U(1)$ transfers universally, SU(2) is forced by $S^3 = \mathrm{SU}(2)$, and SU(3) is gauge-natural but matter-obstructed. The cumulative reading is that the natural gauge content of the construction saturates at $U(1)\times\mathrm{SU}(2)\times\mathrm{SU}(3)$—the structural reach, not a derivation of the Standard-Model gauge group, and consistent with the spectral-triple gauge analysis of Paper 32 [9].

Construction	Count	Tier reading
Scalar graph + scalar photon	1/8	I: graph-native (Gaunt)
Native Dirac graph (scalar photon)	4/8	+ spinor labels
Vector photon + scalar electrons	7/8	II: +6 at $1/(4\pi)$/loop
Vector photon + Dirac electrons	8/8	III: + Furry, free

TABLE I. The 1+6+1 partition of the eight QED selection rules by recoverability on the GeoVac construction. The word "scalar" is overloaded: the 1/8 row is the scalar *photon* on the scalar graph; the 7/8 row is scalar *electrons* with the *vector* photon.

d. The Dirac Rule-B graph and 3D compact $U(1)$ *(Paper 41).* Paper 25's scalar Wilson $U(1)$ has a structural limitation: on the scalar Fock graph the gauge theory factorizes into disjoint per-ℓ two-dimensional grids, so it is really 2D compact $U(1)$ with no genuine three-dimensional phase structure. The cross-ℓ Dirac "Rule B" graph (the $E1$ dipole adjacency, $|\Delta\ell| = 1$) repairs this: every edge flips ℓ-parity, the graph is a single connected component, and its spectral dimension climbs $1.86 \to 2.27 \to 2.54$ toward 3 as $n_{\max} = 3 \to 5$. Wilson $U(1)$ on Rule B is then tested against seven independent structural witnesses (spectral dimension, Migdal–Kadanoff RG flow, Gaussian perimeter law, strong-coupling area law, a DeGrand–Toussaint monopole-density signature decaying over five decades, full Monte-Carlo Wilson loops, and finite-T Polyakov loops on $G_B \times C_{N_t}$). Six pass cleanly and one (full-MC Wilson loops) is a "nuanced weak pass." The verdict is stated at exactly the tier the evidence supports: the theory is *structurally compatible* with 3D compact $U(1)$ (the permanent-confinement, no-finite-T-deconfinement universality class) and inconsistent with 4D—**explicitly not** "uniquely identified with 3D up to universality," and with the $n_{\max} \to \infty$ continuum limit left uncomputed.

III. GRAPH-NATIVE QED: SELECTION RULES AND PERIODS

The QED arc asks how much of perturbative quantum electrodynamics is visible on the bare combinatorial graph, before any continuum projection. Two complementary answers emerge: a clean partition of the selection rules, and a clean accounting of the perturbative transcendentals.

a. The 1+6+1 *partition of selection rules (Paper 33).* The eight standard perturbative-QED selection rules partition into three tiers by *which* ingredient of the construction is needed to recover each (Table I).

On the scalar Fock graph with a scalar (1-cochain) photon, exactly *one* rule survives—the Gaunt/Clebsch–Gordan angular-momentum sparsity baked into the discrete edge set (graph-native, Tier I, free). Promoting the photon to a vector harmonic carrying explicit angular labels (q, m_q) with Wigner-$3j$ vertex coupling recovers *six* angular-momentum rules at once, at a cost of exactly $1/(4\pi)$ per loop—identified as the S^2 Weyl exchange constant of the Hopf base, an entry in Paper 18's calibration tier [4] (Tier II). The eighth rule, charge conjugation / Furry, is recovered for *Dirac* electrons at zero cost by a single-particle spinor phase constraint $V(a, a, q, m_q) = 0$ (proven, sympy-exact for six cases), a kinematic identity distinct from second-quantized C-symmetry (Tier III). The load-bearing structural statement is the dividing line: the photon must carry angular quantum numbers (L, M_L), not merely a spin index at the vertex, to recover the six angular rules—which is precisely why those six sit on the calibration side and only the Gaunt rule is graph-native.

b. The perturbative period tower (Paper 28). The S^3 Dirac spectral geometry (Camporesi–Higuchi spectrum $|\lambda_n| = n + \frac{3}{2}$, Hodge-1 photon, SO(4) vertex rules) produces the *same* transcendentals as flat-space perturbative QED through a purely spectral mechanism organized by operator order and vertex topology. Three results anchor the arc, at theorem/proposition tier (the realized-depth tower is a Proposition):

- **T9 (two-term theorem).** The D^2 spectral zeta is a two-term π^{even} polynomial with rational coefficients at every integer s—no odd-zeta content; operator order (even vs. odd s) is the primary transcendental discriminant.

- **χ_{-4} bridge (Theorem 3).** A spectral-zeta-to-Dirichlet-L identity, $D_{\text{even}}(s) - D_{\text{odd}}(s) = 2^{s-1}\big(\beta(s) - \beta(s-2)\big)$ with $\beta(s) = L(s, \chi_{-4})$, in which vertex parity acts as the Dirichlet character; proven, and verified symbolically and by PSLQ to 100 digits. It is consistent with but does not enforce the Riemann hypothesis for β.

- **Realized-depth tower.** Loop order bounds the multiple-zeta depth but does not automatically mint new irreducibles; the realized depth grows strictly more slowly, $\leq k - 1$. The two-loop sunset constant reduces entirely, $S_{\min} \in \mathbb{Q}[\pi^2, \ln 2, \zeta(3), \zeta(5)]$ (depth ≤ 1); the three-loop chain $S^{(3)} = 31.5726\ldots$ first introduces a genuine depth-2 period, $\zeta(5, 3)$, as its *only* irreducible beyond the single-zeta ring.

On the finite graph, all of these quantities are π-free rationals or algebraic numbers—the transcendentals enter only through the infinite spectral sums of the continuum projection. The continuum self-energy zero is correspondingly *broken* on the graph (the pendant-edge proposition gives $\Sigma_{\text{graph}}(\text{GS}) = 2(n_{\max} - 1)/n_{\max} \to 2 \neq 0$), and is recovered by either of two routes: the vector-photon angular labels of Paper 33, or the plaquette (co-exact) transverse propagator of Papers 28/30. A spectral-sum anomalous moment $F_2/[\alpha/2\pi] = 1.084$ lands within 0.4% of the Parker–Toms curved-space value—reported as a numerical match, not a derivation.

IV. THE FINE-STRUCTURE-CONSTANT OBSERVATION

The arc's most precise—and most carefully bounded—result is Paper 2's. The fine-structure constant satisfies the depressed cubic

$$\alpha^3 - K\alpha + 1 = 0, \qquad K = \pi(B + F - \Delta) = \pi\left(42 + \frac{\pi^2}{6} - \frac{1}{40}\right), \tag{2}$$

where the three coefficients are spectral invariants of the Hopf bundle components: $B = 42$ is the degeneracy-weighted Casimir trace of the S^2 base (with the selection principle $B/N = \dim S^3 = 3$ proven algebraically to fix the cutoff $n_{\max} = 3$ uniquely), $F = \zeta(2) = \pi^2/6$ is the spectral zeta of the S^1 fiber (equivalently the Fock-degeneracy Dirichlet series at the packing exponent $d_{\max} = 4$ [1]), and $\Delta = 1/40$ is an S^3 boundary term equal to the reciprocal of the single-level Dirac degeneracy $g_3^{\text{Dirac}} = 2(n+1)(n+2)|_{n=3} = 40$. The physical root gives $\alpha^{-1} = 137.036011$, matching CODATA to a relative error of 8.8×10^{-8} with zero free parameters; a combinatorial search over 1.92×10^9 comparable formulas returns a p-value of 5.2×10^{-9}.

The status of Eq. (2) must be stated exactly, and it is a **hard prohibition** of the framework that it be stated no more strongly. The combination rule $K = \pi(B + F - \Delta)$ is an **Observation**: an empirical numerical coincidence with structural support, *never* a derivation, conjecture, prediction, or theorem. Each ingredient has an *independently derived* spectral home—a "three-homes" theorem establishes that B, F, Δ live in two distinct sectors of S^3 (scalar and spinor), are three different object types (finite trace, infinite Dirichlet series, single-level degeneracy), and that no spectral construction on S^3 generates all three at once—but their *combination* is not derived. A multi-phase structural-decomposition program eliminated twelve single-mechanism derivations of the rule, leaving the cross-sector sum a genuine open question: in the paper's own assessment, the geometric arena (Link 1) and the per-component invariants (Link 2) are largely established, while the additive combination (Link 3) is "not established." The synthesis inherits this verdict verbatim: what is structural here is each spectral home; what is observational is that they sum to α^{-1}/π.

V. BOUND-STATE PRECISION QED

Paper 36 turns the Dirac-S^3 infrastructure onto Lamb-shift-class observables, where the answer is known independently and the test is one of internal consistency rather than prediction. The hydrogen $2S_{1/2}$–$2P_{1/2}$ Lamb shift comes out at 1052.19 MHz against the experimental 1057.845 MHz—a -0.534% one-loop residual—using only the bare graph plus three standard projections (Fock, the Camporesi–Higuchi spinor lift, and the Drake–Swainson asymptotic subtraction, which is Paper 34's thirteenth catalogued projection [11]), with *no* GeoVac-specific fitted parameter. Companion systems land comparably: muonic hydrogen at -0.10% and the muonium $2S$–$2P$ Lamb shift at $+0.013\%$ (the cleanest match in the set).

The honest reading is that this is *observation-side* physics in the two-layer sense. The framework genuinely *derives* the spectral-sum structure—the Bethe logarithms, computed natively in the acceleration form, agree with Drake–Swainson to better than 1%—but *imports* the QED constants (the 10/9 one-loop self-energy coefficient, the Uehling kernel) from the standard Eides convention; in the Paper 34 language, Layer 1 (the bare Dirac-S^3 spectrum) is π-free, and every π enters through a Layer-2 projection. The two-loop sector is the explicit boundary: the bare iterated spectral action reproduces the flat-space two-loop *divergence* faithfully but does not generate renormalization counterterms, so finite multi-loop predictions remain external input. The result is therefore positioned as one-loop closure with the two-loop closure descoped to a renormalization extension—a precision test passed, not a new QED calculation.

VI. THE SPECTRAL-ACTION GRAVITY SKETCH

The most speculative paper in the arc, Paper 51, asks whether the Fock-projected S^3 supports a sensible gravity reading at the Chamseddine–Connes spectral-action level [17]. It is positioned, correctly and throughout, as an *exploratory, structural-sketch* program—"with the standard caveats of the Connes–Chamseddine program"—not as established physics, and it is unusually candid about what failed.

One result is solid. A spectral-zeta identity on the Camporesi–Higuchi spinor spectrum, $\zeta_{\mathrm{unit}}(-k) = 0$ for all $k \geq 0$ (proven via a Bernoulli identity $B_{2k+1}(3/2) = (2k+1)/4^k$), *forces* the spectral action on S^3 to be *exactly two-term*—Einstein–Hilbert plus a cosmological constant, with no higher-curvature corrections at any order. This is a genuine theorem, and it supplies the algebraic mechanism behind the "remarkable cancellations" noted by Chamseddine–Connes; it is spinor-bundle-specific (the scalar and transverse-traceless sectors retain the full infinite Seeley–DeWitt series). From it follow, at the spectral-action level, a Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A\Lambda^2/(12\pi)$ (continuum replica derivation), an effective Newton constant $G_{\mathrm{eff}} = 6\pi/\Lambda^2$, a de Sitter vacuum, a proven Fierz–Pauli "J-blindness" theorem, and a proven (Layer-2, rate-level) replica-weight-harmless theorem.

The negatives are the point, and they are stated plainly:

- the conical-defect tip coefficient $A = 1/(24\pi)$ was *supplied analytically*, not extracted—all three discrete-substrate discretization schemes undershoot the target;
- the Möbius factor $\alpha/(2\alpha-1)$, once read as a continuum $\alpha > 1$ "mechanism," is a *finite-lattice-spacing artifact*: as the radial mesh refines, the recovery climbs back toward the plain Sommerfeld–Cheeger value, so there is no exotic mechanism to find;
- no cutoff function cures the infrared over-count across all Λ (clean negative);
- the Newton-constant "factor of 2" is a Wald-forced bookkeeping artifact, *not* a scalar-vs-Dirac cone-coefficient difference (2D Dirac and scalar share the same $(1/12)(1/\alpha - \alpha)$ cone term);
- Bisognano–Wichmann wedge entanglement entropy is *not* an area law—it scales as $\sim 2\log n_{\max}$ (area-law fit rejected at $R^2 = 0.83$), so the entropy comes from the replica/spectral-action route, not from wedge entanglement.

The overall verdict is that GeoVac fixes the *structural skeleton* of gravity at the spectral-action level—the two-term action is forced—while the cutoff function remains external calibration data and the full discrete-substrate re-derivation of S_{BH} is an explicitly open research program. Represented at its true tier, this is a proven two-term-exactness theorem wrapped in an honest, still-open exploratory arc.

VII. HONEST SCOPE AND POSITIONING

a. What is robust. The load-bearing results are structural and proven. The Hodge/Wilson dictionary ($L_0 = BB^\top$ matter, $L_1 = B^\top B$ photon) is exact, as is the machine-precision SU(2) maximal-torus reduction that makes Paper 25 the Cartan sector of Paper 30. The QED selection-rule partition 1+6+1 and the perturbative period results—the T9 two-term theorem, the χ_{-4} Dirichlet-L bridge, and the realized-depth tower with $\zeta(5,3)$ as the first genuine depth-2 period—are theorems with symbolic/PSLQ backing. The spectral-action two-term exactness on S^3 is a proven theorem. Each of the three α-ingredients B, F, Δ has an independently derived spectral home.

b. What is conditional or observational. The fine-structure combination rule $K = \pi(B + F - \Delta)$ is an **Observation**, not a derivation—the single most important tier statement in the arc. The bound-state QED matches are observation-side: the spectral-sum structure is derived but the QED constants are imported, and every π enters through a Layer-2 projection. The Rule-B gauge theory is "structurally compatible with 3D compact $U(1)$," not "identified" with it, and its spectral-dimension-to-3 evidence rests on a finite-size extrapolation. The entire gravity sector is a spectral-action-level structural sketch with an open discrete-substrate program.

c. What was tried and did not work. The arc is bounded by clean negatives. SU(3) has no natural matter-coupled lattice gauge theory—the Clebsch–Gordan intertwiner obstruction caps the natural gauge reach (a face of the Coulomb/HO asymmetry). The scalar graph photon misses seven of the eight selection rules because it carries no angular quantum numbers; recovering them is a calibration cost $(1/(4\pi)$ per loop), not graph-native content. Twelve single-mechanism derivations of the α combination rule were eliminated. The gravity sector contributes five recorded negatives—the Möbius finite-spacing artifact, the failed IR-over-count cure, the analytically-supplied (not extracted) tip coefficient, the Wald bookkeeping factor of 2, and the non-area-law wedge entropy—each kept on the record so it is not re-derived.

d. Positioning. Read as a whole, the Group 5 arc is a structural map of how far the spectral geometry of S^3 reaches into electromagnetism, the gauge sector, and gravity, with the boundary between forced structure and free coupling drawn explicitly at every step. The gauge group structure, the selection-rule partition, the QED period tower, and the two-term gravity action are the structural assets; the fine-structure constant, the QED multi-loop constants, and the gravity cutoff are the calibration side. The discrete and continuous descriptions are dual, with no ontological priority asserted for either. The arc is not a derivation of α, not a derivation of the Standard-Model gauge group, and not a theory of quantum gravity; it is a disciplined account of which structures the framework forces and which numbers it must be told.

[1] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section," GeoVac Paper 0 (2026).

[2] J. Loutey, "The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration," GeoVac Paper 2 (2026). [The combination rule $K = \pi(B + F - \Delta)$ is recorded strictly as an Observation; see §IV.]

[3] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[4] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026).

[5] J. Loutey, "The Bargmann–Segal Lattice: π-Free Discretization of the Harmonic Oscillator on S^5," GeoVac Paper 24 (2026).

[6] J. Loutey, "The Hopf Graph as a Lattice Gauge Structure: A Framework Observation," GeoVac Paper 25 (2026).

[7] J. Loutey, "QED on S^3: Spectral Geometry of Perturbative Transcendentals," GeoVac Paper 28 (2026).

[8] J. Loutey, "SU(2) Wilson Lattice Gauge Structure on the GeoVac Hopf Graph: The Non-Abelian Sibling of Paper 25," GeoVac Paper 30 (2026).

[9] J. Loutey, "The GeoVac Spectral Triple," GeoVac Paper 32 (2026).

[10] J. Loutey, "The 1+6+1 Partition of QED Selection Rules on S^3: Graph Topology, Angular Momentum, and the Dirac Spinor Phase Constraint," GeoVac Paper 33 (2026).

[11] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (2026).

[12] J. Loutey, "Bound-State QED on S^3: One-Loop Closure of the Hydrogen Lamb Shift," GeoVac Paper 36 (2026).

[13] J. Loutey, "State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples on S^3," GeoVac Paper 38 (2026).

[14] J. Loutey, "Wilson $U(1)$ Lattice Gauge on the Dirac-Rule-B Graph: Structural Compatibility with 3D Compact $U(1)$ on a Non-Cubic Substrate, Including the Polyakov Monopole-Density Signature," GeoVac Paper 41 (2026).

[15] J. Loutey, "Gravity from the GeoVac Spectral Action: Continuum Results and the Discrete-Substrate Program," GeoVac Paper 51 (2026).

[16] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," J. Geom. Phys. **75**, 71 (2014); arXiv:1301.3480.

[17] A. H. Chamseddine and A. Connes, "The Spectral Action Principle," Commun. Math. Phys. **186**, 731 (1997).